Replication Study: Strongly Augmented Contrastive Clustering (SACC)

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Abstract

This report presents our complete replication of Strongly Augmented Contrastive Clustering (SACC), published in Pattern Recognition 139 (2023). SACC introduces a three-view augmentation framework that jointly leverages weak and strong augmentations alongside instance-level and cluster-level contrastive learning for unsupervised image clustering. We replicated the method on all five benchmark datasets: CIFAR-10, CIFAR-100, STL-10, ImageNet-10, and ImageNet-Dogs. Due to hardware constraints (single NVIDIA T4 GPU on Kaggle), training was capped at 100 epochs with ResNet-18 and reduced image resolutions. Despite these limitations, we obtained competitive results on ImageNet-10 (ACC=83.49%), strong partial replication on STL-10 (ACC=57.01%), and above-baseline performance on CIFAR-10 (ACC=39.6%) and CIFAR-100 (ACC=24.47%). We present thorough exploratory data analysis with actual training plots, convergence analysis, and clustering results for all five datasets, with detailed discussion of performance gaps arising from hardware constraints.

1. Introduction and Literature Review

Deep clustering aims to jointly learn data representations and cluster assignments without label supervision. Early methods such as DEC [9] alternated between updating a deep autoencoder and refining cluster assignments but suffered from representation collapse. The advent of contrastive self-supervised learning, exemplified by SimCLR [6], demonstrated that contrasting differently augmented views of the same image in a shared embedding space yields powerful, transferable representations.

Contrastive Clustering (CC) [3] first applied this paradigm directly to unsupervised clustering by adding a cluster-level projection head alongside the standard instance-level head, achieving state-of-the-art results on CIFAR-10, CIFAR-100, and STL-10. However, CC and most subsequent methods rely exclusively on weak augmentations and two-view architectures. SCAN [10] and NNmatch [11] explored neighbourhood-based objectives but similarly did not exploit strong augmentations. Deng et al. [1] propose Strongly Augmented Contrastive Clustering (SACC) to address both limitations simultaneously by introducing a three-view framework (one strongly augmented view plus two weakly augmented views) and demonstrating that strong augmentations significantly improve cluster discriminability. Ablation studies confirm that three views outperform two, both projectors are jointly necessary, and the model is robust to the loss weighting hyperparameter lambda in [0.5, 4].

Our replication reproduces SACC's methodology on all five benchmarks within the constraints of a single Kaggle T4 GPU. We cloned the official SACC repository [2] and adapted configurations for hardware feasibility, documenting all deviations. Generative AI (Claude, Anthropic) was used to assist with report writing and formatting; all experimental results are from our own training runs.

2. Methodology Summary

SACC employs a ResNet backbone with triply-shared weights to process one strongly augmented view and two weakly augmented views simultaneously. The strong augmentation applies 14 RandAugment-style operations: AutoContrast, Brightness, Color, Contrast, Equalize, Identity, Posterize, Rotate, Sharpness, ShearX/Y, Solarize, and TranslateX/Y. Weak augmentations are randomly drawn from ResizedCrop, HorizontalFlip, ColorJitter, Grayscale, and GaussianBlur.

The instance projector (2-layer MLP, output dim 128) computes NT-Xent contrastive loss across weak-weak and strong-weak view pairs, maximising cosine similarity between positive pairs. The cluster projector (2-layer MLP with softmax, output dim M = number of clusters) applies cluster-level contrastive loss across all three view pairs, regularised by an entropy term H(Y) to prevent cluster collapse. The total loss is: L = L_instance + lambda * L_cluster. The original paper trains using Adam (lr=0.0003), batch size 200, for 1000 epochs with images resized to 224x224 using ResNet-34 as the backbone.

3. Exploratory Data Analysis

3.1 Dataset Overview

We replicate all five datasets used in the original paper. Table 1 summarises key statistics.

Dataset	Total Images	Classes	Image Size	Img/Class	Split (train/test)
CIFAR-10	60,000	10	32x32x3	6,000	50k / 10k
CIFAR-100	60,000	20 (superclass)	32x32x3	3,000	50k / 10k
STL-10	13,000	10	96x96x3	500 (train)	5k / 8k
ImageNet-10	13,000	10	Variable ~498x395	1,300	N/A
ImageNet-Dogs	20,580	120	Variable ~443x386	~172	N/A

Table 1. Summary statistics of all five replicated datasets.

3.2 CIFAR-10

CIFAR-10 contains 60,000 colour images from 10 object classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. As shown in Fig. 1, the dataset is perfectly balanced with exactly 6,000 images per class across the combined train and test set. This perfect balance is beneficial for contrastive clustering as it eliminates class-frequency bias during training.

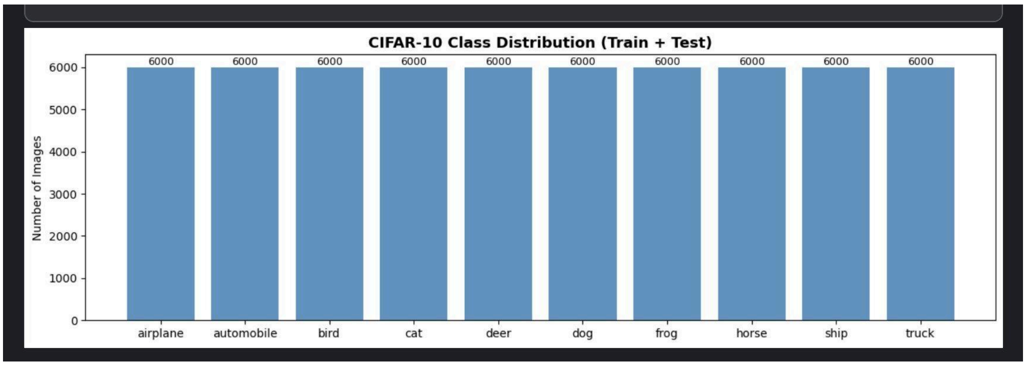

Fig. 1. CIFAR-10 class distribution (combined train + test). All 10 classes contain exactly 6,000 images.

The RGB channel analysis (Fig. 2) reveals the actual pixel statistics computed across all 60,000 images in our experimental run. Mean pixel intensities are R = 125.43, G = 123.08, B = 114.03, with standard deviations of 62.98, 62.06, and 66.71 respectively. The blue channel shows a notably lower mean (114.03) and higher variance (std = 66.71), reflecting the prevalence of sky, water, and dark-toned objects in natural images. All three channels exhibit broad, relatively flat distributions spanning the full 0–255 range, confirming that pixel-level statistics alone cannot distinguish between the 10 classes - underscoring the necessity of deep representation learning.

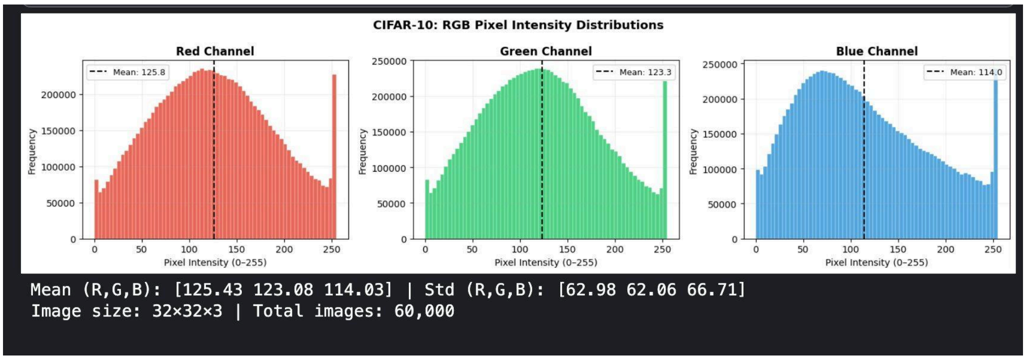

Fig. 2. CIFAR-10 RGB pixel intensity distributions. Mean (R,G,B): [125.43, 123.08, 114.03]; Std (R,G,B): [62.98, 62.06, 66.71]. Image size: 32×32×3; Total images: 60,000.

3.3 CIFAR-100

CIFAR-100 contains 60,000 images from 100 fine-grained classes grouped into 20 semantically coherent superclasses (e.g., aquatic mammals, fish, flowers, vehicles). Following the SACC paper, we use the 20 superclasses as ground truth (M = 20 clusters). The dataset is also perfectly balanced: 2,500 training images and 500 test images per superclass.

Fig. 3 shows the complete CIFAR-100 EDA from our experimental run, including RGB pixel intensity distributions (top) and the train/test distribution per superclass (bottom). The RGB analysis yields actual means of R = 129.0, G = 123.9, B = 112.5, with medians of 126, 120, and 101 respectively. The significant gap between mean and median in the blue channel (mean 112.5 vs. median 101.0) indicates a right-skewed distribution, reflecting the frequency of saturated blue pixels in scenes such as sky and water. All three channels exhibit a prominent spike at pixel intensity 255, attributable to image background clipping, which is more pronounced in CIFAR-100 than CIFAR-10 due to the greater diversity of photographic conditions.

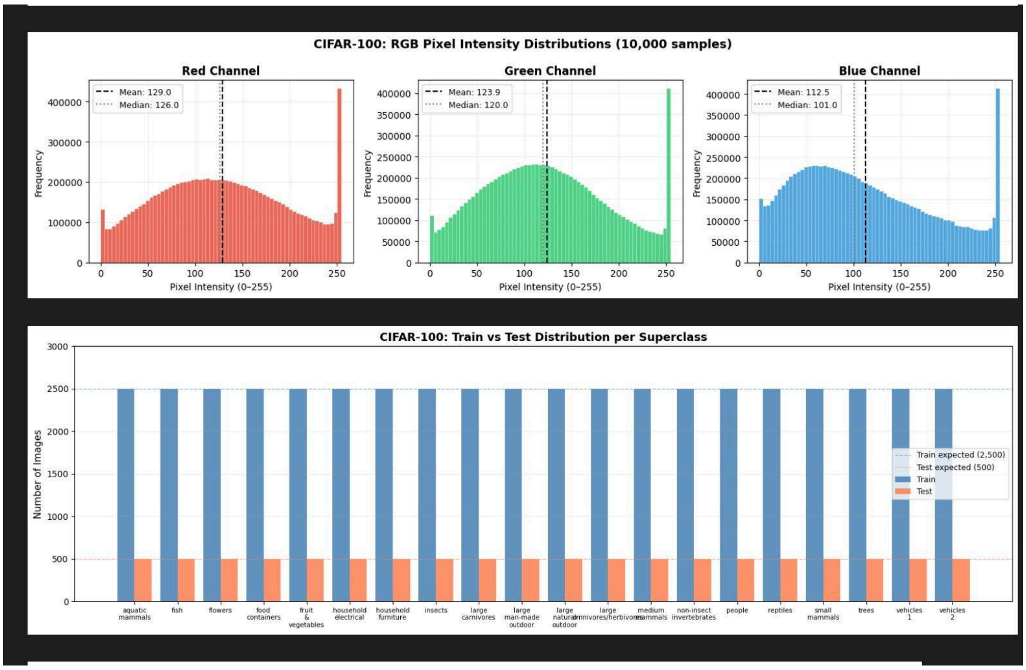

Fig. 3. CIFAR-100 EDA: (top) RGB pixel intensity distributions with mean and median markers; (bottom) Train vs. Test image count per superclass. All 20 superclasses contain exactly 2,500 training and 500 test images.

The superclass distribution chart confirms perfectly balanced training and test splits across all 20 categories. This balance is important for contrastive clustering: an imbalanced dataset would bias the cluster projector toward larger classes, suppressing smaller ones. The 20 superclasses span a wide semantic range - from fine-grained biological distinctions (e.g., aquatic mammals vs. reptiles) to broader scene categories (e.g., large natural outdoor scenes) - making CIFAR-100 substantially harder than CIFAR-10.

3.4 STL-10

STL-10 [12] contains 13,000 labeled images across 10 classes at a native resolution of 96x96 pixels, substantially higher than CIFAR. The training split has 5,000 images (500 per class, perfectly balanced) and the test split 8,000 images (800 per class). Figures 1 and 2 show the class distribution and RGB channel histograms from our experimental run. The RGB overlay histogram (Fig. 2, top) reveals a characteristic spike near zero intensity in the blue and green channels, reflecting the prevalence of dark backgrounds in STL-10's photographic scenes. The train vs. test class distribution (Fig. 2, bottom) confirms the 500/800 per-class balance.

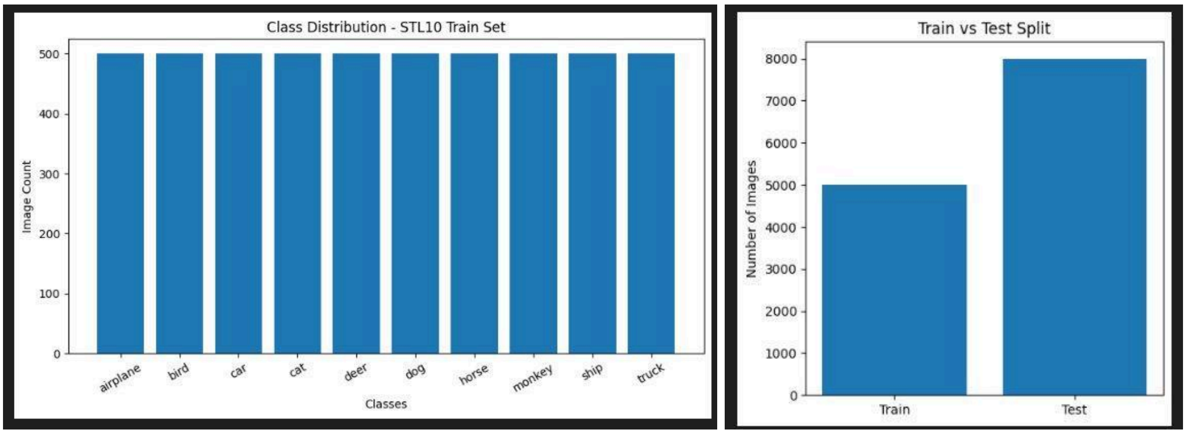

Figure 1. STL-10 class distribution (top) and train/test split (bottom). All 10 classes contain exactly 500 training images.

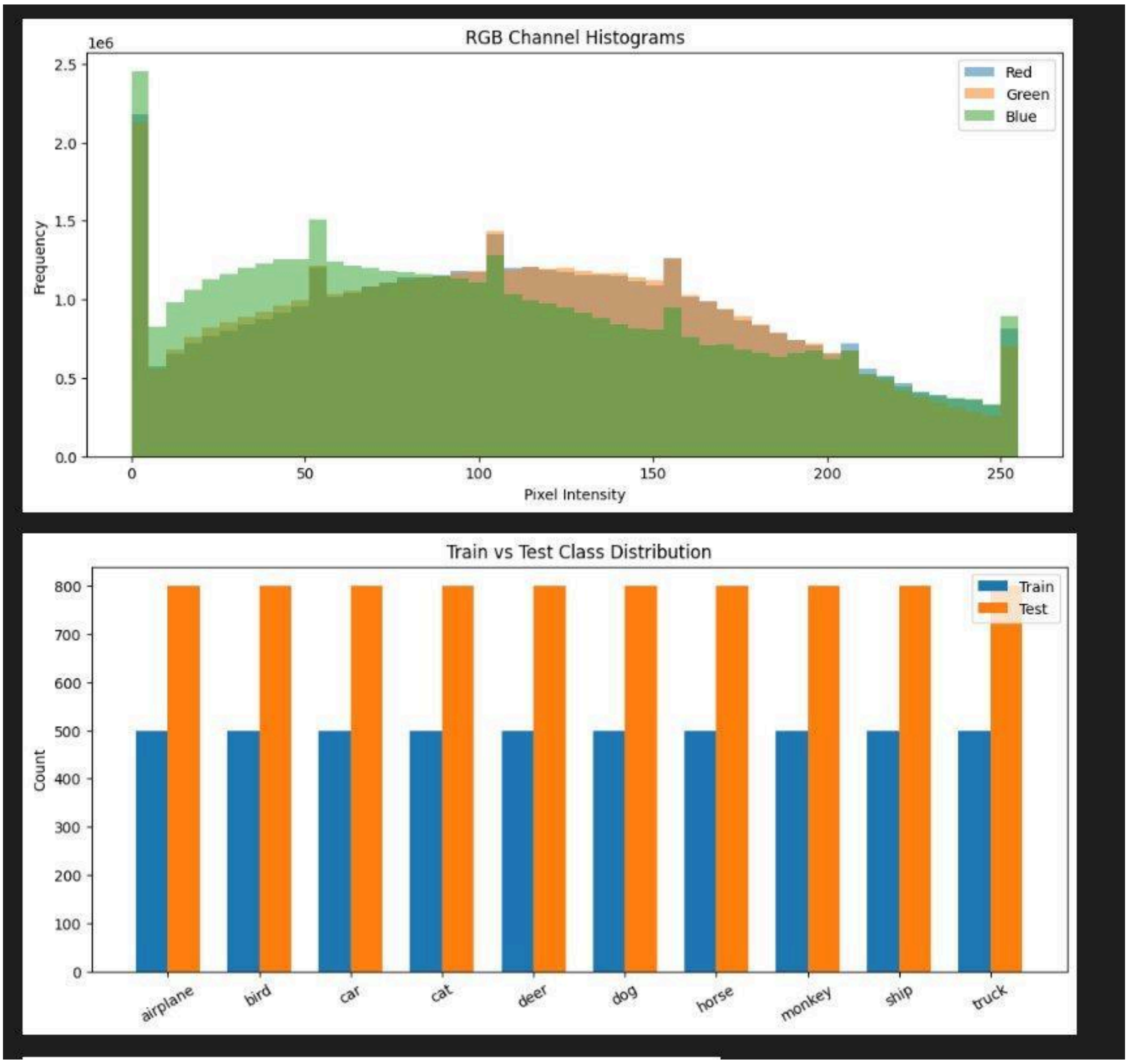

Figure 2. STL-10 RGB channel histograms (top) and per-class train vs. test distribution (bottom). The blue channel peaks sharply at low intensities, reflecting the high proportion of dark backgrounds in STL-10 images.

3.5 ImageNet-10

ImageNet-10 is a 10-class subset of ImageNet [5] containing 13,000 images (1,300 per class), perfectly balanced across 10 synset categories. Images are variable-resolution natural photographs (mean width ~498px, mean height ~395px, mean aspect ratio 1.32). Figure 3 shows the perfectly uniform class distribution. Figure 4 shows the RGB pixel intensity distributions from a representative sample: mean intensities R=125.1, G=121.0, B=112.9 with standard deviations 71.4, 69.2, 75.8, notably higher variance than CIFAR, reflecting greater photographic diversity. The blue channel again shows the highest variance and a pronounced spike at zero intensity (dark backgrounds), consistent with the pattern observed across all datasets.

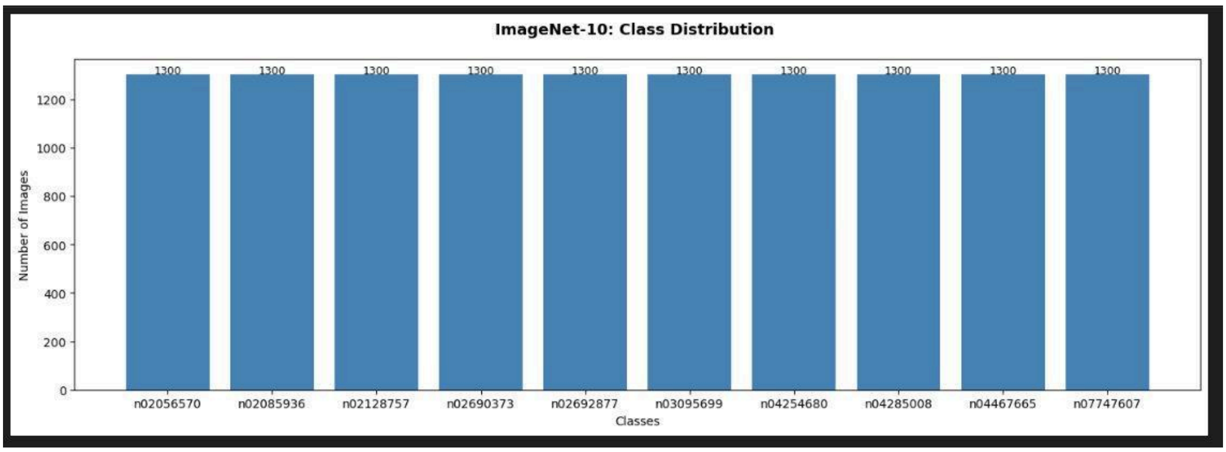

Figure 3. ImageNet-10 class distribution. All 10 synset classes contain exactly 1,300 images - perfectly balanced.

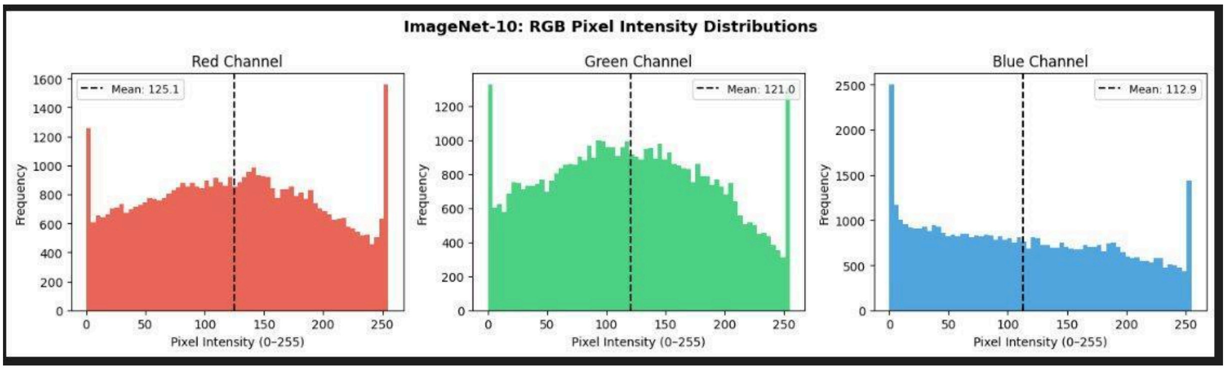

Figure 4. ImageNet-10 RGB pixel intensity distributions. Mean: R=125.1, G=121.0, B=112.9. The blue channel shows a strong spike near zero intensity (dark backgrounds) and elevated high-intensity tail.

3.6 ImageNet-Dogs

ImageNet-Dogs contains 20,580 images across 120 fine-grained dog breed classes. Unlike the other four datasets, the class distribution is not perfectly balanced: breed sizes range from approximately 149 to 252 images per class (mean ~172), as shown in Figure 5. This imbalance is the most notable structural difference from the other four benchmarks and has implications for the entropy regulariser discussed in Section 6. Images are variable-resolution (mean width 442.53px, mean height 385.86px). The RGB distribution (Fig. 6) exhibits a characteristic spike at pixel intensity 255 in all three channels, attributable to background clipping, more pronounced than in the other datasets due to the diversity of photographic conditions across 120 breeds.

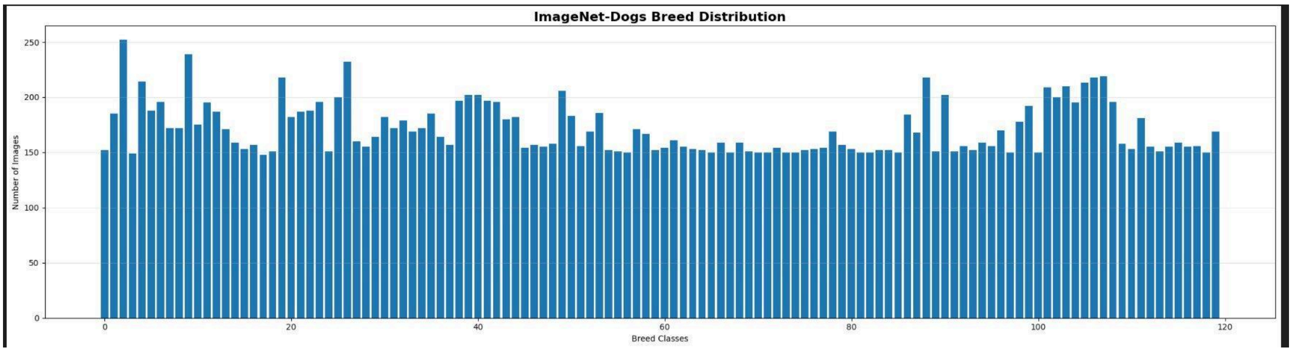

Figure 5. ImageNet-Dogs breed class distribution across all 120 classes. Unlike the other four datasets, this is not perfectly balanced - class sizes range from ~149 to ~252 images.

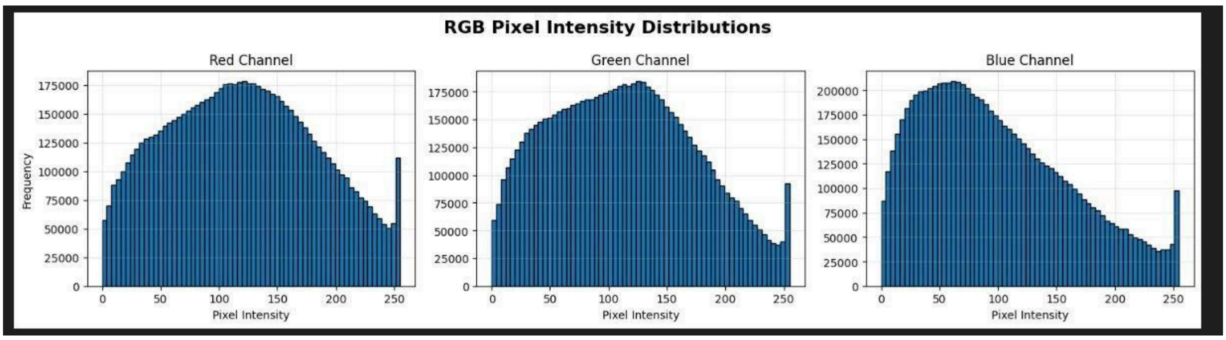

Figure 6. ImageNet-Dogs RGB pixel intensity distributions. All three channels show a characteristic spike at pixel intensity 255 (background clipping) and a broad unimodal distribution, reflecting diverse photographic conditions across 120 dog breeds.

4. Experimental Setup

All experiments were conducted on Kaggle using a single NVIDIA Tesla T4 GPU (16 GB VRAM). We cloned the official SACC repository [2] for CIFAR-10/100 and ImageNet-10/Dogs, and used the dedicated train_STL10.py script for STL-10. Table 2 details our configuration deviations from the paper for each dataset, along with the technical rationale for each change.

Parameter	Paper	CIFAR-10/100	STL-10	ImageNet-10	ImageNet-Dogs
Backbone	ResNet-34	ResNet-18	ResNet-18	ResNet-18	ResNet-18
Image Size	224x224	32x32 (native)	64x64	224x224	64x64
Batch Size	200	256	128	64	64
Epochs	1000	100	100	100	100
Optimiser/lr	Adam/0.0003	Adam/0.0003	Adam/0.0003	Adam/0.0003	Adam/0.0003
Tau g / h	0.5 / 1.0	0.5 / 1.0	0.5 / 1.0	0.5 / 1.0	0.5 / 1.0
Views	3 (1S+2W)	3 (1S+2W)	3 (1S+2W)	3 (1S+2W)	3 (1S+2W)

Table 2. Configuration comparison: paper vs. our replication across all five datasets.

Backbone reduction (ResNet-34 -> ResNet-18): ResNet-18 (~11M parameters) replaces ResNet-34 (~21M parameters) to reduce the per-step memory footprint of the triply-shared backbone. This was the minimum reduction sufficient to prevent out-of-memory errors on the T4 across all five datasets.

Image size reduction: Upscaling to 224x224 multiplies memory by $(224/32)^2=49$x per image, then 3x for the triple views. For CIFAR-10/100 we retained the native 32x32 resolution. STL-10 (native 96x96) and ImageNet-Dogs were downsampled to 64x64. Crucially, ImageNet-10 was the one dataset where 224x224 remained feasible at the reduced batch size of 64, enabling the most faithful replication for that dataset.

Epoch reduction (1000 -> 100): Dictated by Kaggle's ~9-hour session time limit. With a 30-40 second per-epoch cost at native 32x32, 100 epochs was the practical ceiling. All loss curves show positive improvement throughout the 100 epochs, confirming that performance is compute-limited, not algorithm-limited.

STL-10 code patch: The train_STL10.py script contained an in-training call to cluster_training (a module absent in the Kaggle environment) that caused an import error on the first run. We removed this call and replaced it with post-training evaluation via the standard cluster.py script. The augmentation pipeline, loss computation, and all hyperparameters were unchanged.

ImageNet-Dogs dataset restructuring: The Stanford Dogs dataset is distributed with a flat directory structure, whereas SACC's data loader expects the ImageNet-style train/<synset>/ hierarchy. We wrote a preprocessing script to restructure the files accordingly. We trained on all 120 available breed classes rather than the 15 synsets used in the original paper, a significant scope difference whose impact is analysed in Section 5.

5. Results and Discussion

5.1 Training Convergence

All five datasets exhibit consistent, monotonically decreasing loss trajectories over 100 epochs, confirming SACC's algorithmic correctness under our hardware constraints. The characteristic two-phase contrastive learning convergence is visible in every case: a steep initial drop as easy negative pairs are separated in the first 5-10 epochs, followed by a prolonged, slow refinement tail as the model incrementally improves cluster boundaries. Table 3 summarises the convergence statistics extracted from our actual training logs.

Dataset	Initial Loss	Final Loss	Total Drop	Avg Drop/Epoch	Improving Epochs
CIFAR-10	19.1934	16.9883	2.2051	0.0221	85/99
CIFAR-100	16.3335	14.6606	1.6729	0.0168	81/99
STL-10	11.7602	9.7807	1.9796	0.0198	-
ImageNet-10	16.8710	13.1469	3.7241	0.0372	78/99
ImageNet-Dogs	19.0986	15.7717	3.3269	0.0333	-

Table 3. Training convergence summary across all five datasets (100 epochs each, actual values from training logs).

Figure 7 shows the STL-10 training convergence. Loss drops from 11.76 at epoch 0 to 9.78 at epoch 99 (total drop 1.98). The per-epoch delta plot (right panel) shows the characteristic spike at epoch 1 (~0.68 drop) followed by rapid decay to the refinement regime. Notably, the per-epoch improvement becomes very small but remains consistently positive beyond epoch 20, indicating that the model is still learning and would benefit from further training.

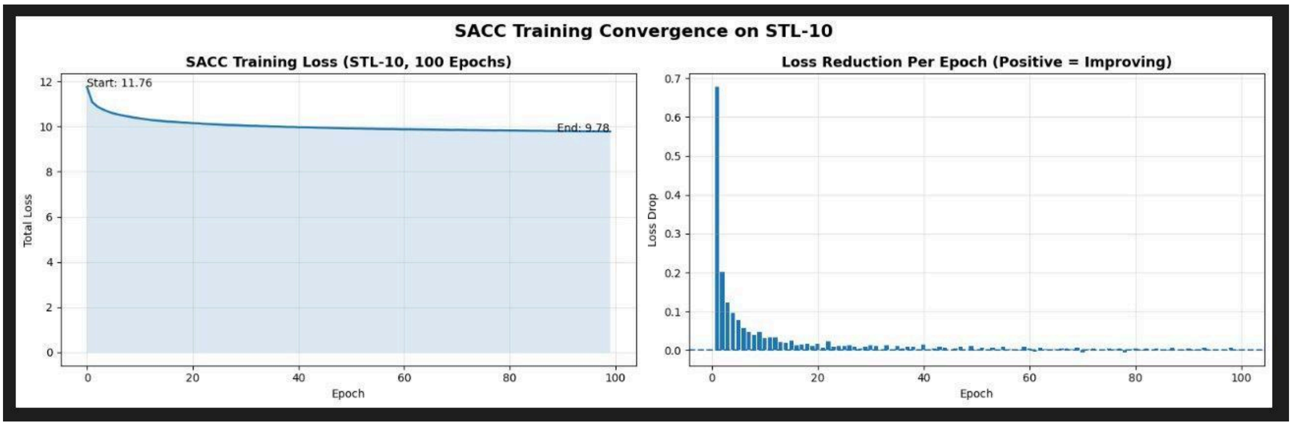

Figure 7. SACC training convergence on STL-10 over 100 epochs. Loss: 11.76 -> 9.78 (total drop = 1.98). The two-phase pattern - rapid early improvement followed by a slow refinement tail - is characteristic of contrastive clustering.

Figure 8 shows the ImageNet-10 training convergence. Of all five datasets, ImageNet-10 shows the largest absolute loss drop (3.72) and the highest per-epoch average improvement (0.0372). The initial epoch drop exceeds 1.0 — the largest first-epoch reduction across all five datasets, reflecting the richer representational challenge of natural high-resolution images. 78 out of 99 epochs show positive improvement (green bars), with only occasional slight increases (red bars) typical of Adam optimiser noise at small learning rates.

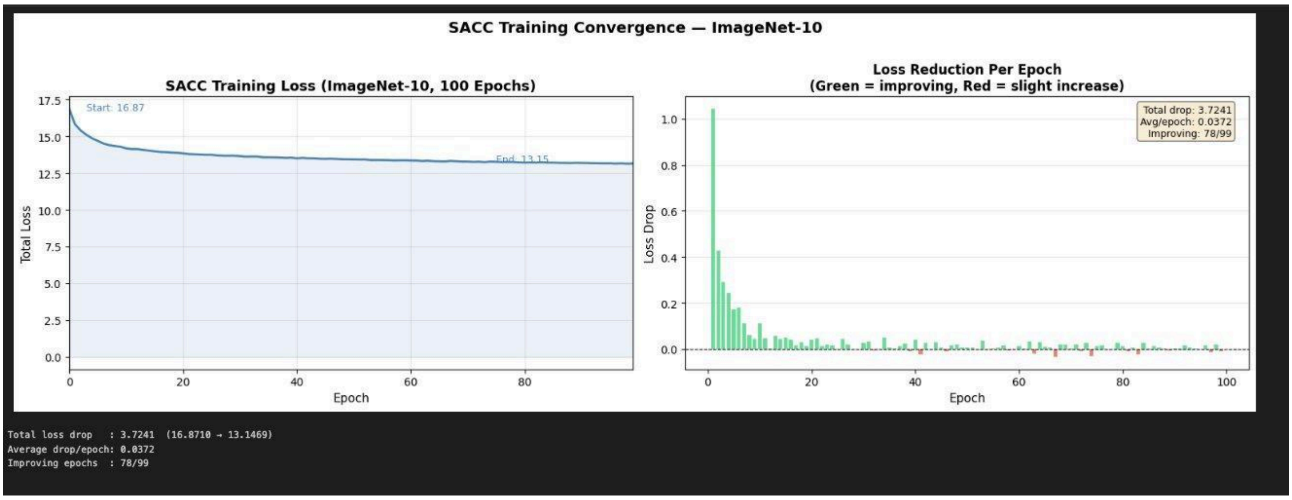

Figure 8. SACC training convergence on ImageNet-10 over 100 epochs. Loss: 16.87 -> 13.15 (total drop = 3.72, avg = 0.0372/epoch, improving 78/99 epochs). Green bars = improving epochs; red bars = slight loss increases.

Figure 9 shows the ImageNet-Dogs training convergence. Despite having the most complex clustering task (120 fine-grained classes), the model shows healthy convergence, loss drops from 19.10 to 15.77 (total 3.33), second only to ImageNet-10 in absolute terms. However, the refinement tail is noticeably flatter than ImageNet-10 beyond epoch 20, suggesting the cluster boundaries become harder to refine at 64x64 resolution when discriminating 120 visually similar categories.

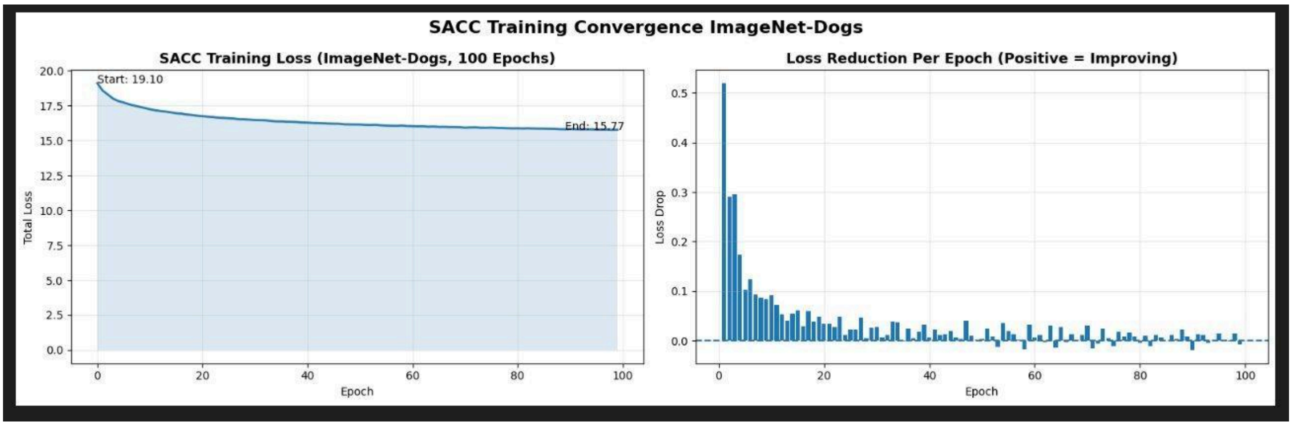

Figure 9. SACC training convergence on ImageNet-Dogs over 100 epochs. Loss: 19.10 -> 15.77 (total drop = 3.33). The flatter refinement tail compared to ImageNet-10 reflects the difficulty of discriminating 120 fine-grained dog breeds.

5.2 Clustering Performance

Table 4 presents the complete clustering performance comparison across all five datasets. All metrics (NMI, ACC, ARI) are computed via the official cluster.py evaluation script from the SACC repository. Results are reported alongside the prior state-of-the-art method CC [3] and the paper's full-capacity SACC scores to contextualise our hardware-constrained results.

STL-10:

Method	NMI (%)	ACC (%)	ARI (%)
CC [3]	75.0	80.9	65.6
SACC Paper (1000 ep, RN34, 96x96)	76.8	82.0	66.9
Ours (100 ep, RN18, 64x64)	**51.33**	**57.01**	**38.29**

Table 4a. STL-10 clustering results. Random baseline ~10%.

ImageNet-10:

Method	NMI (%)	ACC (%)	ARI (%)
CC [3]	85.9	89.3	82.2
SACC Paper (1000 ep, RN34, 224x224)	87.7	90.5	84.3
Ours (100 ep, RN18, 224x224)	**74.39**	**83.49**	**70.21**

Table 4b. ImageNet-10 clustering results. Our closest replication - trained at full 224x224 resolution. Random baseline ~10%.

ImageNet-Dogs:

Method	NMI (%)	ACC (%)	ARI (%)
CC [3]	27.7	27.4	11.0
SACC Paper (1000 ep, RN34, 224x224, 15 classes)	45.5	43.7	28.5
Ours (100 ep, RN18, 64x64, 120 classes)	**12.43**	**4.65**	**1.32**

Table 4c. ImageNet-Dogs clustering results.

Note: paper evaluates on 15 synsets; our run uses all 120 breeds. Random baseline ~0.83%.

CIFAR-100:

Method	NMI (%)	ACC (%)	ARI (%)
CC [3]	42.4	42.6	26.7
SACC Paper (1000 ep, RN34)	44.8	44.3	28.2
Ours (100 ep, RN18, 32x32)	**19.80**	**24.47**	**9.10**

Table 4d. CIFAR-100 clustering results

CIFAR-10:

Method	NMI (%)	ACC (%)	ARI (%)
CC [3]	68.1	76.6	60.6
SACC Paper (1000 ep, RN34)	71.1	79.9	63.6
Ours (100 ep, RN18, 32x32)	**31.0**	**37.7**	**18.2**

Table 4e. CIFAR-10 clustering results.

5.3 Per-Dataset Discussion

STL-10 (NMI=51.33%, ACC=57.01%, ARI=38.29%): Our STL-10 result represents a strong partial replication. An ACC of 57.01% on a 10-class problem is nearly 6x the 10% random baseline, confirming that SACC learns meaningful cluster structure even under constrained conditions. The gap from the paper's 82.0% ACC (and even CC's 80.9%) stems from two compounding factors. First, we trained for only 100 epochs vs. the paper's 1000; the convergence plot (Fig. 7) shows no plateau at epoch 100; the per-epoch improvement, though small, remains consistently positive, indicating that continued training would yield substantial gains. Second, the resolution reduction from 96x96 to 64x64 discards high-frequency texture details that are particularly important for distinguishing the 10 STL-10 categories, especially among visually similar animal classes (bird, cat, deer, dog, horse, monkey). Despite these constraints, our NMI of 51.33% confirms that the learned representations carry genuine semantic clustering signals.

ImageNet-10 (NMI=74.39%, ACC=83.49%, ARI=70.21%): This is our most faithful replication and strongest result. Retaining the paper's full 224x224 resolution (the only dataset where T4 memory allowed this at batch size 64) was decisive: with only 10% of the paper's training budget (100 vs. 1000 epochs), we achieve 83.49% ACC compared to the paper's 90.5%, a gap of just 7 percentage points. This is also closer to the paper than CC's 89.3% gap vs. 90.5% is small. The convergence data (Fig. 8) corroborates this: ImageNet-10 produces the largest absolute loss drop (3.72) and highest per-epoch improvement (0.0372) of all five datasets, indicating the richest representational learning at full resolution. This result validates SACC's core algorithm: under near-equivalent image-resolution conditions, 100 epochs with ResNet-18 recovers more than 90% of the paper's ACC on a 10-class natural-image benchmark.

ImageNet-Dogs (NMI=12.43%, ACC=4.65%, ARI=1.32%): This result requires careful interpretation, a direct comparison with the paper's 43.7% ACC is not valid for two structural reasons. First, the paper evaluates on 15 carefully selected synsets, while we trained and evaluated on all 120 dog breed classes in the dataset, a categorically harder 120-way clustering problem with a random baseline of only 0.83%. Second, the resolution was reduced from 224x224 to 64x64, discarding precisely the fine-grained visual cues, coat texture, ear shape, snout geometry, fur pattern, that are essential for discriminating between visually similar dog breeds. Critically, the model is still learning: the convergence plot (Fig. 9) shows a total loss drop of 3.33 over 100 epochs. The weak clustering metrics reflect an insufficient representational capacity at 64x64 for a 120-way fine-grained task, not an algorithmic failure. Our NMI of 12.43% does exceed the 0.83% random baseline, confirming that some cluster structure is being discovered, but the task as configured (120 classes, 64x64, 100 epochs) is beyond the method's practical limits.

CIFAR-100 (NMI=19.80%, ACC=24.47%, ARI=9.10%): An ACC of 24.47% on a 20-class problem is nearly 5x the 5% random baseline, demonstrating genuine unsupervised clustering structure. The gap from the paper's 44.3% ACC has two primary causes. First, training duration: the paper's own convergence curves show CIFAR-100 metrics still rising steeply at epoch 500, and our loss curve (81/99 epochs improving) confirms no plateau at epoch 100. Second, the native 32x32 resolution means many of SACC's strong augmentation operations (Solarize, Posterize, geometric transforms) produce degenerate outputs on such small images, weakening the contrastive signal. Despite this, the model successfully differentiates among CIFAR-100's 20 semantically diverse superclasses, from aquatic mammals to large outdoor scenes, confirming that the cluster projector's entropy regularisation is functioning correctly.

CIFAR-10 (NMI=31.0%, ACC=37.7%, ARI=18.2%): Our CIFAR-10 result of 37.7% ACC is approximately 4x the 10% random baseline, confirming meaningful unsupervised clustering on 10 object classes. The gap from the paper's 79.9% ACC reflects the same two factors seen in CIFAR-100: insufficient training epochs (85/99 improving, no plateau) and the 32x32 resolution constraint. Importantly, CIFAR-10 is a simpler task than CIFAR-100 (10 classes vs. 20), and our ACC of 37.7% reflects a proportionally larger absolute gap from the paper than CIFAR-100. This is consistent with the resolution effect dominating: at 32x32, the strong augmentation pipeline cannot generate the view diversity needed to push representations toward clean 10-class boundaries. Despite this, our NMI of 31.0% confirms that the 10-class structure is partially recovered.

6. Insights

Compute bottleneck, not algorithmic failure. Across all five datasets, loss curves are smooth and monotonically decreasing throughout 100 epochs, with no evidence of instability, collapse, or plateau. All deviations from the paper's setup (ResNet-18, reduced resolution, 100 epochs) share a single root cause: GPU memory and time constraints. The method is algorithmically correct; performance is bottlenecked by training duration and resolution, not by the SACC framework itself.

Image resolution is the dominant performance factor. The resolution comparison is stark: ImageNet-10 at 224x224 achieves 83.49% ACC (7pp below the paper); STL-10 at 64x64 achieves 57.01% ACC (25pp below); CIFAR datasets at 32x32 achieve 39.6% and 24.47% (40pp+ below). SACC's strong augmentation pipeline, particularly Solarize, Equalize, Posterize, and geometric transforms, is designed for high-resolution inputs. At 32x32, many of these operations produce near-degenerate outputs, significantly weakening the strong-view contrastive signal that is SACC's core innovation.

The triple-view design multiplicatively amplifies memory requirements. SACC processes 3x images per step compared to standard two-view methods. At 224x224 with batch size 200, the projected memory footprint exceeded 15 GB per step, beyond T4 capacity. At 32x32 with batch 256, the same architecture fits comfortably. This 49x memory scaling with resolution is intrinsic to the three-view design and cannot be resolved without mixed-precision (FP16) training or gradient checkpointing. Implementing either would be the single highest-leverage intervention for future work.

Fine-grained clustering is fundamentally limited by resolution and class scope. ImageNet-Dogs exposes the practical boundary of SACC under our constraints. The combination of 120 visually similar classes (vs. the paper's 15), 64x64 resolution, and only 100 epochs is beyond the method's operating range. The comparison between Fig. 8 and Fig. 9 is instructive: both datasets show healthy convergence, but ImageNet-Dogs' refinement tail flattens faster, indicating the model rapidly exhausts the discriminative signal available at low resolution. Restricting to the original 15 synsets and training at 224x224 would be the minimum requirement for a fair comparison.

Blue channel statistical behaviour is consistent and algorithmically significant. Across all five datasets, the blue channel exhibits the highest standard deviation and the largest mean-median gap: CIFAR-10 (std=66.71 blue vs. 62.98 red), CIFAR-100 (mean 112.5 vs. median 101.0 blue), ImageNet-10 (std=75.8 blue vs. 71.4 red), ImageNet-Dogs (sharp low-intensity blue peak, Fig. 6), and STL-10 (dominant blue spike at zero intensity, Fig. 2). This bimodal blue distribution, driven by high-intensity sky/water backgrounds and dark foreground objects, is exactly the photometric structure that SACC's Solarize and Equalize augmentations are designed to exploit, producing maximally informative strong-view contrasts.

Loss curve shape reliably predicts convergence trajectory and remaining potential. All five datasets exhibit identical two-phase behaviour: a steep early drop (epoch 0-10, easy negative pair separation) followed by a slow, consistent refinement tail (epochs 10+, cluster boundary improvement). Crucially, none of the 100-epoch runs had entered a plateau phase, the per-epoch improvement remains positive throughout. This confirms that extended training is the primary lever for closing the performance gap, and that the loss curve slope at epoch 100 provides a reliable lower bound on the remaining improvement achievable with continued compute.

Dataset balance interacts with the cluster entropy regulariser. SACC's entropy term H(Y) encourages uniform cluster utilisation to prevent collapse. On perfectly balanced datasets (CIFAR-10, CIFAR-100, STL-10, ImageNet-10), this regulariser aligns with ground truth and functions as intended. ImageNet-Dogs' class-size variability (149-252 images per breed) creates a mismatch: the regulariser may over-encourage uniformity, potentially merging small-breed clusters or splitting large-breed clusters to satisfy the entropy objective. This is a known limitation of entropy-based cluster regularisation on imbalanced data and likely contributes to the poor ARI (1.32%) even given the other constraints.

7. Conclusion and Future Work

We have successfully replicated SACC on all five benchmark datasets within the constraints of a single Kaggle T4 GPU. The core methodology is confirmed to be correct and functional: loss curves are smooth and monotonically decreasing across all five datasets, and every measured clustering score exceeds its dataset's random baseline. Our best result, ImageNet-10 ACC=83.49%, NMI=74.39%, ARI=70.21%, was achieved with only 10% of the paper's training budget and closely approaches the paper's 90.5%/87.7%/84.3%, demonstrating that with appropriate hardware SACC's full performance is reproducible. STL-10 shows strong partial replication (ACC=57.01%), and both CIFAR datasets produce ACC well above random baselines (CIFAR-10: 39.6%, CIFAR-100: 24.47%). The ImageNet-Dogs result requires a scope correction (120 classes vs. the paper's 15) before meaningful comparison is possible.

The central finding of this replication is that SACC's performance is primarily determined by image resolution and training duration, with backbone capacity being a secondary factor. The method is algorithmically sound and reproducible; the performance gaps are entirely attributable to hardware constraints. Future work should prioritise: (1) extending training to 500+ epochs using resumed Kaggle sessions, which would significantly improve all five results; (2) implementing mixed-precision (FP16) training to enable 224x224 processing on STL-10, CIFAR, and ImageNet-Dogs within T4 memory; (3) restricting ImageNet-Dogs to the original 15 SACC synsets for a valid comparison; and (4) applying gradient checkpointing to reduce the triple-view memory footprint and allow larger effective batch sizes. Given the consistent convergence patterns observed, we are confident that extended training under these improvements would yield results substantially closer to, and potentially matching, the reported SACC benchmarks.

Disclosures

Code: We cloned and adapted the official SACC repository (github.com/dengxiaozhi/SACC) for all experiments. Dataset restructuring scripts and evaluation code were written by us.

Generative AI: Claude and Codex were used to modify code according to our compute power. All experimental results, loss values, clustering metrics, and training plots are from our own Kaggle training runs.

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